Analysis of Continuous Real-time Data

Yunyi Ren

UCD, Department of Public Health Sciences

Background

- A review of the study section taught by Kaizer, El al. Colorado school of public health, in 2024 CSP
- GitHub Page for detailed information
- Data collected by wearable device such as smartwatch or biometric sensor
- Data will be summarized in desired form:
  - One record per person per day
  - One record per person per event
  - One record per person per epoch

What did they talk about?

- General introduction
- Why choose this approach (situation, advantages)
- Importance of organize data files and tips
- Define events based on criteria (clinical decision)
- Summary and visualizing data
- Modeling and interpretation

Common Wearables Research Questions

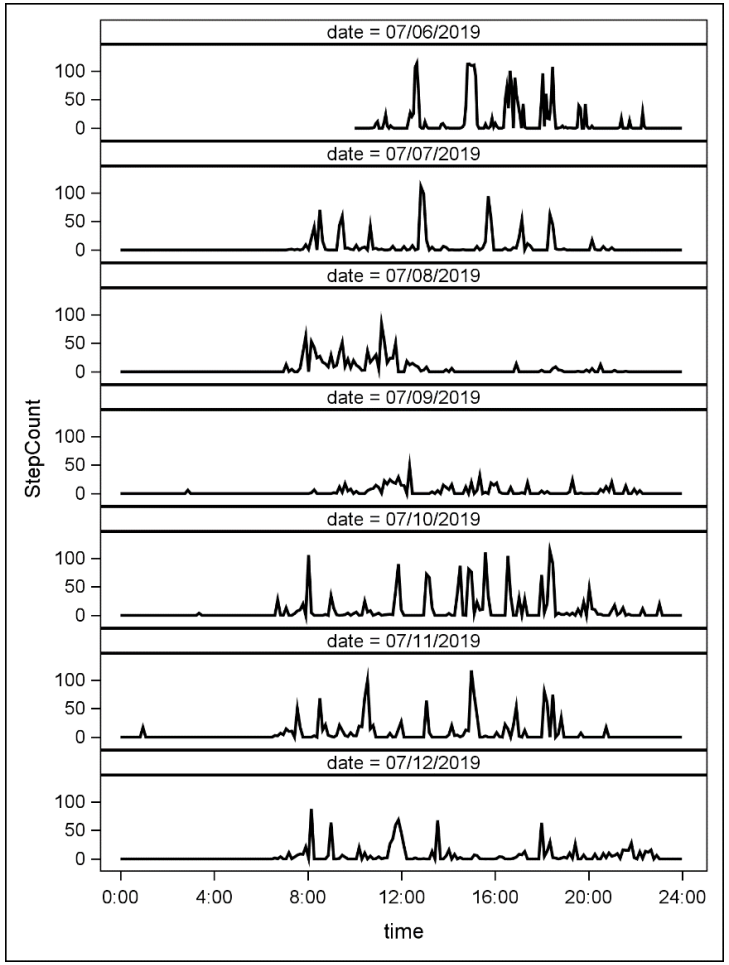
- Does the outcome change?
- Are there group differences in the outcome?
- How does the outcome change over time?
- How is change in one outcome related to change in a second outcome?

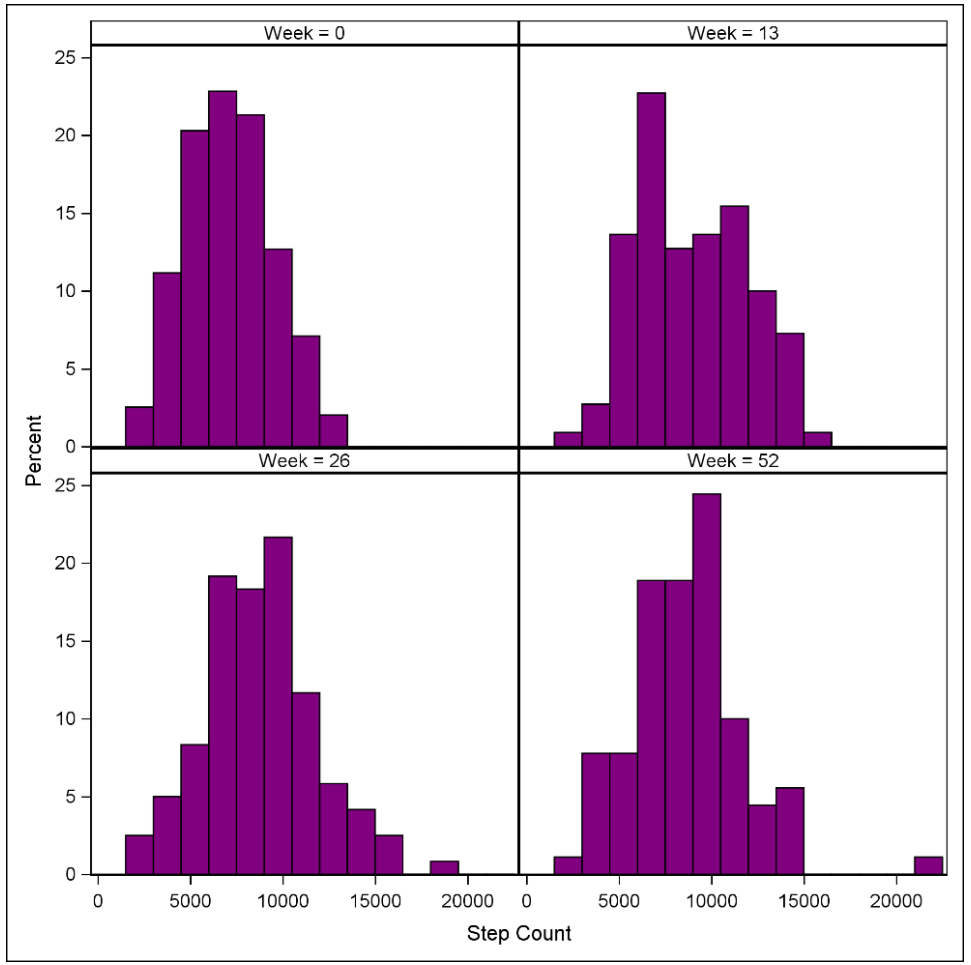
Example - Summary

Change over time

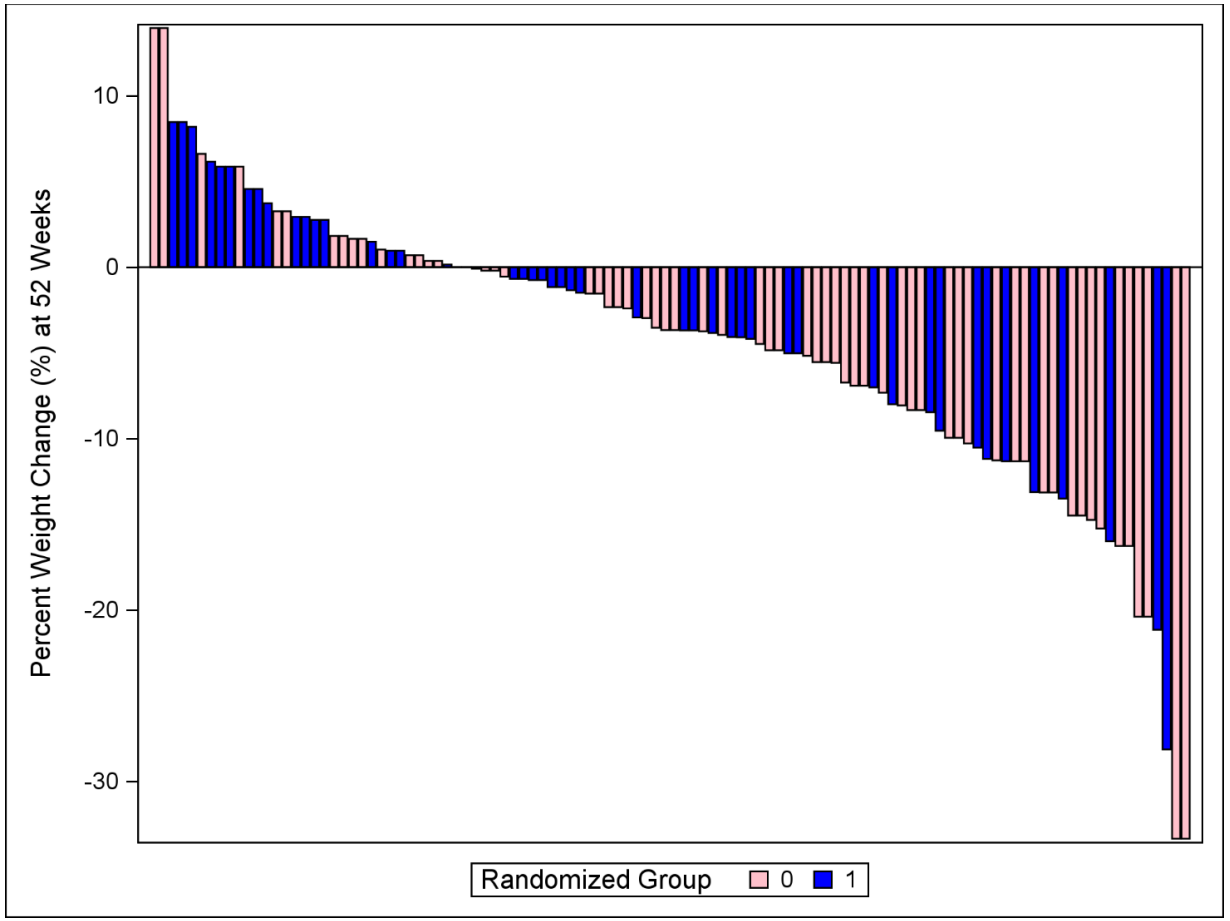
Measure	*Week*	*All*	*Change from baseline*	*Randomized Group 1*	*Change from baseline*	*Randomized Group 0*	*Change from baseline*
Step Count	0	7134.3 (2313.0)		7394.4 (2378.1)		6924.4 (2248.1)	
	13	8940.9 (2996.4)	1520.5 (2519.2)	9405.0 (3006.3)	1649.7 (2434.8)	8643.1 (2974.4)	1437.6 (2586.6)
	26	8841.5 (3011.8)	1347.8 (2381.9)	9303.6 (3029.9)	1449.4 (2647.1)	8564.3 (2986.7)	1286.8 (2224.1)
	52	8615.5 (3074.0)	1111.2 (2834.5)	9466.0 (3031.4)	1447.7 (3011.4)	8099.1 (3009.8)	906.9 (2729.1)
Weight (kg)	0	99.2 (15.8)		97.2 (15.6)		100.9 (15.8)	
	13	91.9 (15.8)	-6.6 (6.3)	92.4 (15.9)	-4.8 (5.1)	91.5 (15.8)	-8.0 (6.8)
	26	92.7 (15.8)	-5.9 (7.8)	93.2 (14.3)	-3.7 (6.6)	92.4 (16.8)	-7.4 (8.2)
	52	92.4 (16.0)	-4.3 (8.3)	93.9 (14.6)	-2.6 (7.8)	91.3 (17.0)	-5.4 (8.5)
Energy intake (kcals)	0	1703.8 (474.9)		1626.5 (410.6)		1772.1 (517.8)	
	13	1329.7 (383.0)	-383.9 (416.9)	1521.9 (301.7)	-173.8 (301.6)	1182.5 (374.8)	-548.0 (422.5)
	26	1314.0 (411.1)	-421.4 (433.6)	1461.9 (375.7)	-264.1 (452.6)	1212.2 (406.0)	-538.6 (382.3)
	52	1343.1 (442.5)	-385.3 (475.2)	1538.0 (376.1)	-148.7 (469.1)	1205.9 (437.0)	-565.1 (398.0)

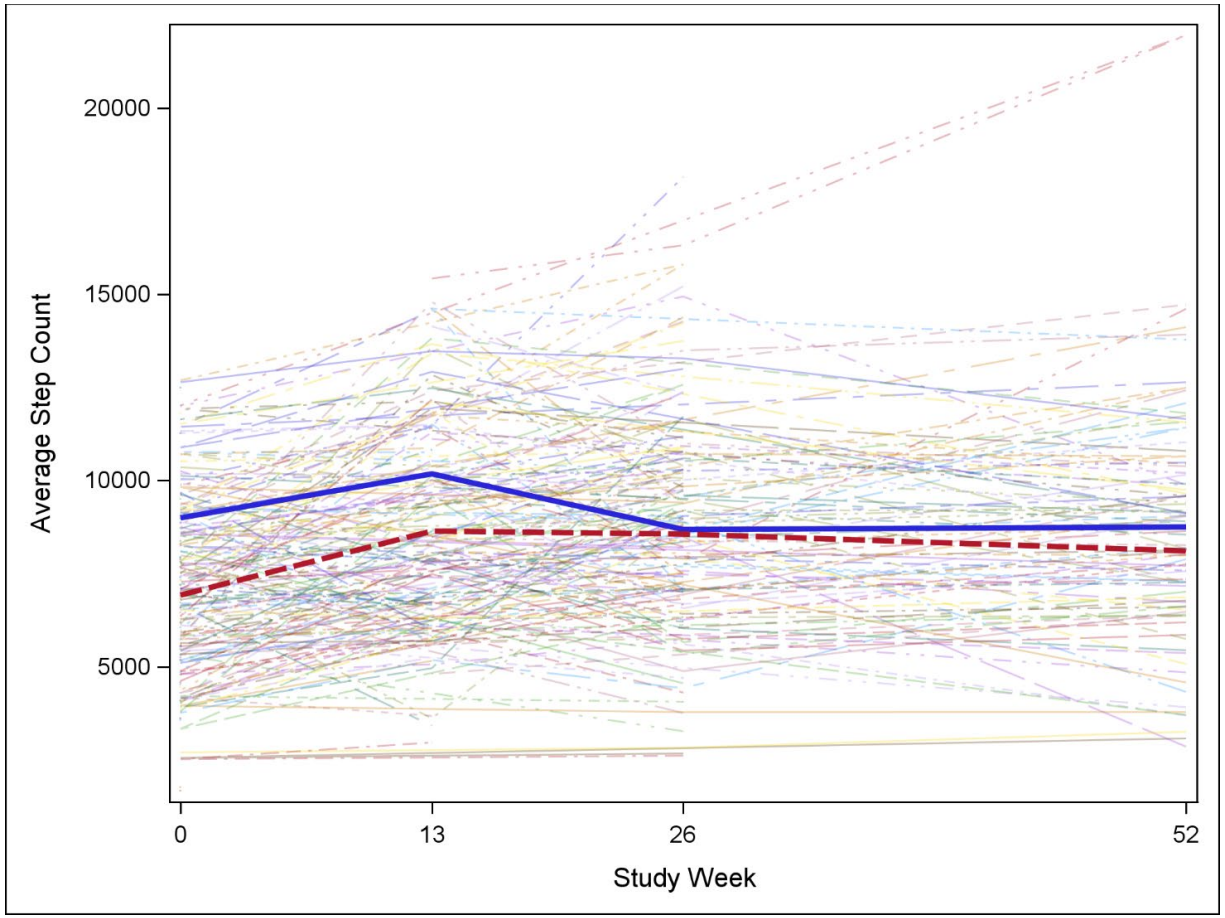
Example - Visualization





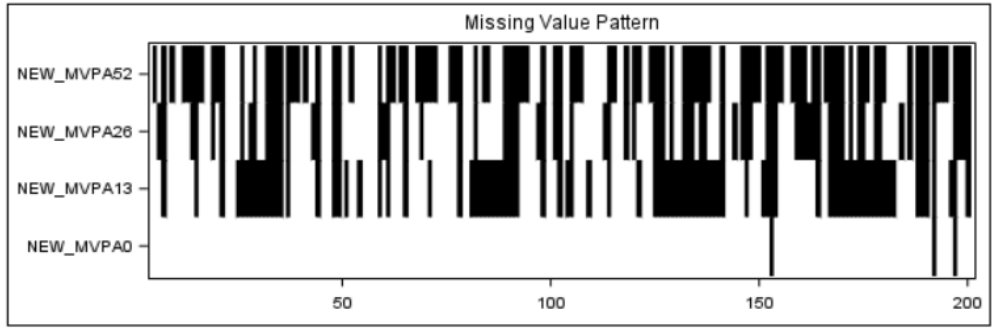
Example - Visualization





Modeling – missing data

- Characterize missingness
  - What
  - How much
  - Why
  - Patterns
- For example, missing on treatment vs control
- Another way is to visualize missing data



Modeling - Method

- Two sample t-test
- GLM
- Repeated measure ANOVA
- GEE
- Mixed Effect Model
- Parallel Process Growth Curve Models

Modeling - Method

- Two sample t-test
- GLM
- Repeated measure ANOVA
- GEE
- Mixed Effect Model
- Parallel Process Growth Curve Models

Do people in the intervention group *increase/decrease* physical activity more than control?

Modeling - Method

- Two sample t-test
- GLM
- Repeated measure ANOVA
- GEE
- Mixed Effect Model
- Parallel Process Growth Curve Models

Is there a between group difference in physical activity at the end of the study, when controlling for baseline physical activity?

Modeling - Method

- Two sample t-test
- GLM
- Repeated measure ANOVA
- GEE
- Mixed Effect Model
- Parallel Process Growth Curve Models

Sometimes the simplicity of RM ANOVA is preferred, when:

- Simple Design
- No missing data (or minimal missingness)
- Continuous, unbounded outcome
- Normal residuals
- No clustering
- Categorical time
- Balanced repeats
- Your research question is whether two groups change differently

Modeling - Method

- Two sample t-test
- GLM
- Repeated measure ANOVA
- GEE
- Mixed Effect Model
- Parallel Process Growth Curve Models

- Account for dependence of repeated measures
- Estimates population averages (no random effects)
- Simple nested or repeated measure designs

Does it fit your question, the data, model assumptions?

Modeling - Method

- Two sample t-test
- GLM
- Repeated measure ANOVA
- GEE
- Mixed Effect Model
- Parallel Process Growth Curve Models

Does the intervention influence how physical activity changes over time?

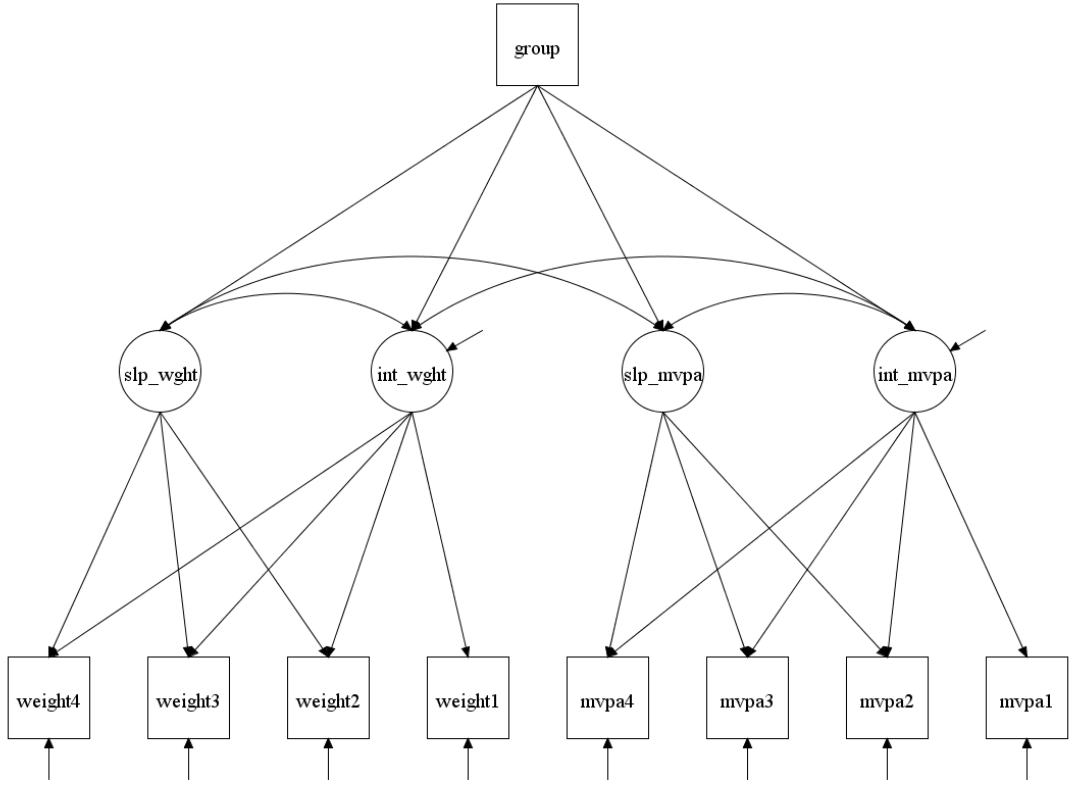
Modeling - Method

- Two sample t-test
- GLM
- Repeated measure ANOVA
- GEE
- Mixed Effect Model
- Parallel Process Growth Curve Models

What is the relationship between the change in physical activity and change in weight?

(Latent) Growth Curves Analysis

Examines how two or more longitudinal processes change over time. These models allow us to explore how the intercept (initial value) and growth rate (slope) of one variable are associated with the parameters of another variable



Result Example

- There was a significant group effect on the growth process of physical activity, b = 4.08, Z = 3.68, p < .001, such that the diet + exercise group experienced faster increases than the diet only group.
- There was not a significant group effect on the growth process of weight.

Weight

Physical Activity

Weight

